

Tema lab 1

4.

- c. limuzine = C_{11}^2
- c. furgonete = C_6^3
- c. cabriolete = C_3^1
- c. far = $C_{11}^2 \cdot C_6^3 \cdot C_3^1$
- c. pos = C_{20}^6

$$P = \frac{c. far}{c. pos} = \frac{C_{11}^2 \cdot C_6^3 \cdot C_3^1}{C_{20}^6} = \frac{55 \cdot 20 \cdot 3}{38760} = \frac{3300}{38760} = 0,0851$$

5. a) $P_{loto} = \frac{C_6^6}{C_{49}^6} = \frac{1}{13983816}$

300 kg = 300 000 000 mg

~~boabe = $\frac{30}{300\,000\,000}$~~

boabe = $\frac{300\,000\,000}{30} = 10\,000\,000$

$P_{boabe} = \frac{1}{10\,000\,000}$

$\Rightarrow P_{boabe} > P_{loto}$

b) $P_{loto} = \frac{C_6^5 \cdot C_{43}^1}{C_{49}^6} = \frac{6 \cdot 43}{13983816} = \frac{258}{13983816} = \frac{1}{54200,83}$

1,5 kg = 1 500 000

boabe = $\frac{1\,500\,000}{30} = 50000$

$P_{boabe} = \frac{1}{50000}$

$\Rightarrow P_{boabe} > P_{loto}$

6.

- c. roata A = C_6^2
- c. roata B = C_5^1
- c. roata c = C_4^3
- c. far = $C_6^2 \cdot C_5^1 \cdot C_4^3$
- c. pos = C_{15}^6

$$P = \frac{c. far}{c. pos} = \frac{C_6^2 \cdot C_5^1 \cdot C_4^3}{C_{15}^6} = \frac{300}{5005} = 0,0599$$

$$8. \quad c. \text{ fav} = C_2^2$$

$$c. \text{ pos} = C_{10}^2$$

$$p = \frac{c. \text{ fav}}{c. \text{ pos}} = \frac{C_2^2}{C_{10}^2} = \frac{1}{45} = 0,0222$$

$$9. a) \quad p = \frac{C_6^6}{C_{49}^6} = \frac{1}{13983816}$$

$$b) \quad p = \frac{C_6^5 \cdot C_{43}^1}{C_{49}^6} = \frac{258}{13983816} = \frac{1}{54200,83}$$

$$c) \quad p = \frac{C_6^4 \cdot C_{43}^2}{C_{49}^6} = \frac{13545}{13983816} = \frac{1}{1032,39}$$

$$d) \quad p = \frac{C_6^3 \cdot C_{43}^3}{C_{49}^6} = \frac{246820}{13983816} = \frac{1}{56,65}$$

10. a) A, B, C 3 evenimente

A singur se realizează: $P(A \cap \bar{B} \cap \bar{C})$

b) A și C se realizează dar nu în B : $P(A \cap \bar{B} \cap C)$

c) A, B, C se realizează: $P(A \cap B \cap C)$

d) cel puțin unul dintre cele 3 evenimente se produce:

$$P(A \cup B \cup C)$$

e) cel puțin două evenimente din cele trei se produc:

$$P((\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C) \cup (A \cap B \cap \bar{C}) \cup (A \cap B \cap C))$$

f) cel mult un eveniment se produce:

$$P((A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C) \cup (\bar{A} \cap \bar{B} \cap \bar{C}))$$

g) niciun eveniment nu se produce: $P(\bar{A} \cap \bar{B} \cap \bar{C})$

h) exact două evenimente se produc:

$$P((A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C))$$

11. 5 extrageri

A_i eveniment, cu $1 \leq i \leq 5$

a) A - numai o bilă este albă

$$A = \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap \overline{A_{i_2}} \cap \overline{A_{i_3}} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}})$$

b) B - cel puțin o bilă este neagră

$$B = \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4} \cap \overline{A_{i_5}}) \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap$$

$$\cap A_{i_3} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}}) \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap \overline{A_{i_3}} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}}) \cup A \cup$$

$$\cup (\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4} \cap \overline{A_5})$$

c) C - obținerea a cel puțin două bile albe

$$C = (\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4} \cap \overline{A_5}) \cup A \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap \overline{A_{i_3}} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}})$$

d) D - obținerea a cel puțin trei bile albe

$$D = (A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5) \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap A_{i_4} \cap \overline{A_{i_5}})$$

$$\bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}})$$

$$\text{sau } D = \overline{C}$$

e) $\bar{E}$ - numai două bile sunt regre

$$\bar{E} = \bigcup_{i_1, i_2, i_3, i_4, i_5} (A_{i_1} \cap A_{i_2} \cap A_{i_3} \cap \overline{A_{i_4}} \cap \overline{A_{i_5}})$$

14. a) A - unul din cei 4 pești prinși este un crap

$$P_A = \frac{C_3^1 \cdot C_7^3}{C_{10}^4} = \frac{105}{210} = 0,5$$

b) B - cel puțin unul din cei 4 pești prinși este un crap

$$P_B = 1 - \frac{C_7^4}{C_{10}^4} = 1 - \frac{35}{210} = 1 - 0,1666 = 0,8334$$

c) C - primul pește prins este un crap

$$P_C = \frac{3}{10} = 0,3$$

d) D - al doilea pește prins este un crap

$$P_D = \frac{3}{10} \cdot \frac{2}{9} + \frac{7}{10} \cdot \frac{3}{9} = \frac{6+21}{90} = \frac{27}{90} = 0,3$$

e) $\bar{E}$ - primii doi pești prinși sunt crați

$$P_{\bar{E}} = \frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} = 0,0666$$

f) F - cel puțin unul din primii doi pești prinși este crap

$$P_F = \frac{C_3^1 \cdot C_7^1}{C_{10}^2} = \frac{21}{45} = 0,4666$$

g) G - fiecare din ultimii trei pești prinși cântărește mai mult decât cel precedent

$$P_G = \left(\frac{1}{2}\right)^3 = 0,1250$$

12. penton rest probl cons:

N - nr de rosete regre

R - nr de rosete rosii

T - nr de rosete in total

c - favorabila = C_R^2

c - totale = C_T^2

$$p = \frac{c \text{ fav}}{c \text{ tot}} = \frac{C_R^2}{C_T^2} = \frac{1}{2}$$

$$\frac{\frac{R(R-1)}{2}}{\frac{T(T-1)}{2}} = \frac{1}{2} \Rightarrow \frac{R(R-1)}{T(T-1)} = \frac{1}{2}$$

$$2R(R-1) = T(T-1)$$

a) $T \rightarrow$ nr min

$$T = R + N \quad \text{in } T \neq \{0, 1\}$$

$$\text{cons } N = 1 \Rightarrow 2R(R-1) = (R+1)(R-1+1)$$

$$2R^2 - 2R = R^2 + R \Rightarrow R^2 - 3R = 0 \Rightarrow R(R-3) = 0 \Rightarrow R = 3$$

$$T = 3 + 1 = 4 \text{ rosete}$$

b) $N = \text{var} \Rightarrow N = 2h$

$$2R(R-1) = (R+2h)(R+2h-1)$$

$$2R^2 - 2R = R^2 + 2Rh - R + 2Rh + 4h^2 - 2h$$

$$R^2 - R - 4Rh - 4h^2 + 2h = 0$$

$$(R - 2h)^2 - R - 2h = 0$$

$$(R - 2h)(R - 2h) - (R - 2h) = 0$$

$$(R - 2h)(R - 2h - 1) = 0$$

$$(R - 2h)(R - 2h - 1) - 8h^2 = 0$$

$$(R - 2h)(R - 2h - 1) = 8h^2$$

$$2 \text{ nr cons} = 8h^2 \Rightarrow \text{cons } h = 1$$

$$h = 2$$

$$h = 3$$

$$(N = 6)$$

$$2 \text{ nr cons imm} = 72/9$$

$$R - 6 = 9 \Rightarrow R = 15 \Rightarrow T = 21$$

13. 10 cartanare

$$\begin{aligned} n_{\text{totale}} &= A_{10}^5 - A_9^4 \\ &= 30240 - 3024 \\ &= 27216 \end{aligned}$$

foloim un script nt a det nr de nr divizibile cu 455
n m de distincte

$$c_{\text{fav}} = 66$$

$$p = \frac{c_{\text{fav}}}{n_{\text{tot}}} = \frac{66}{27216} = 0,0024$$

15. A, B evenimente legate

$$p(A \cup B) = 0,5$$

$$p(A \cup \bar{B}) = 0,7$$

$$p(A \cup B) = p(A) + p(B) - p(A) \cdot p(B)$$

$$p(A \cup \bar{B}) = p(A) + p(\bar{B}) - p(A) \cdot p(\bar{B})$$

$$= p(A) + (1 - p(B)) - p(A) \cdot (1 - p(B))$$

$$= p(A) + 1 - p(B) - p(A) + p(A) \cdot p(B)$$

$$= 1 - p(B) + p(A) \cdot p(B)$$

$$p(A \cup B) + p(A \cup \bar{B}) = p(A) + p(B) - p(A) \cdot p(B) + 1 - p(B) + p(A) \cdot p(B)$$

$$= 1 + p(A) = 0,6 \Rightarrow p(A) = -0,4$$